

Operation & Maintenance Manual

830E-1AC

DUMP TRUCK

SERIAL NUMBERS 830E-1AC A40939-A40961

ENGINE 16V160

This material is proprietary to Komatsu America Corp. and is not to be reproduced, used, or disclosed except in accordance with written authorization from Komatsu America Corp.

It is our policy to improve our products whenever it is possible and practical to do so. We reserve the right to make changes or improvements at any time without incurring any obligation to install such changes on products sold previously.

Due to this continuous program of research and development, revisions may be made to this publication. It is recommended that customers contact their distributor for information on the latest revision.

INTRODUCTION	
FOREWORD	1-2
ABOUT THIS MANUAL	1-3
WARNINGS.....	1-4
California Proposition 65 Warnings	1-4
ALERTS	1-5
TRUCK MODEL ILLUSTRATION	1-6
NON-OEM PARTS IN CRITICAL SYSTEMS	1-7
STANDARD CHARTS AND TABLES	
GENERAL INFORMATION	2-2
Effect Of Special Lubricants	2-2
Suggested Sources For Rust Preventive Grease	2-2
STANDARD TIGHTENING TORQUES	2-3
Class 10.9 Capscrews And Class 10 Nuts	2-3
SAE Grade 5 And Grade 8 Capscrews.....	2-4
SAE Grade 9 Capscrews	2-5
T-Bolt Clamps.....	2-5
Split Flange Clamps	2-6
Flared Tube And Hose Fittings	2-6
JIC 37° Swivel Nuts	2-7
Pipe Thread Fittings.....	2-8
O-Ring Boss Fittings	2-9
O-Ring Face Seal Fittings	2-10
CONVERSION MULTIPLIERS	2-11
Metric To English	2-11
English To Metric	2-12
Temperature Conversions	2-13
GENERAL SAFETY	
GENERAL SAFETY	3-2
Safety Rules.....	3-2
Truck Safety Features	3-2
Clothing And Personal Items	3-2
Unauthorized Modification	3-2
Leaving The Operator's Seat	3-2
Mounting And Dismounting	3-2
Fire Extinguishers And First Aid Kits	3-3
Precautions For High Temperature Fluids.....	3-3
Asbestos Dust Hazard Prevention	3-3
Fire Prevention	3-3
ROPS Precautions.....	3-3
Preventing Injury From Work Equipment	3-4
Precautions For Optional Attachments	3-4
Precautions When Starting The Machine	3-4
PRECAUTIONS FOR TRUCK OPERATION	3-5
Safety Is Thinking Ahead	3-5
Safety At The Worksite	3-5
Preparing For Operation.....	3-5
Ventilation For Enclosed Areas	3-5
Mirrors, Windows, And Lights	3-5
Before Starting The Engine	3-5
Seat Belts And Tethers.....	3-5
Emergency Egress Ladders	3-6
OPERATING THE MACHINE.....	3-7
General Information	3-7

Starting The Engine	3-7
Traveling In The Machine	3-7
Traveling In Reverse	3-8
Traveling On Slopes	3-8
Traveling On Snow	3-8
Traveling Near High Voltage Cables	3-8
Traveling On Loose Ground	3-8
Ensuring Good Visibility	3-8
Avoid Damage To The Dump Body	3-8
Safe Parking Of The Machine	3-9
Loading The Machine	3-9
Dumping The Machine	3-9
Towing The Machine	3-9
EXTERIOR LIGHTING	3-10
Fog Lights	3-10
Headlights	3-10
Brake Lights	3-10
Retard Lights	3-11
Clearance/Turn Signal Lights	3-11
Backup Lights	3-11
Retard Lights	3-11
Brake Lights	3-12
Backup Alarm	3-12
Backup Lamp	3-12
PRECAUTIONS FOR MAINTENANCE	3-13
BEFORE PERFORMING MAINTENANCE	3-13
Electrical System Isolation	3-13
Securing The Dump Body	3-14
DURING MAINTENANCE	3-15
Warning Tag	3-15
Proper Tools	3-15
Attachments	3-15
Working Under The Machine	3-15
Keeping The Machine Clean	3-16
Adding Fuel Or Oil To The Machine	3-16
Radiator Coolant Level	3-16
Use Of Lighting	3-16
Handling High Pressure Hoses	3-16
Precautions With High Pressure Oil	3-16
Maintenance Near High Temperatures And High Pressures	3-17
Rotating Fan And Belts	3-17
Waste Materials	3-17
Use Of Tie-Off Anchor During Maintenance	3-17
Handling Tires	3-17
Tire Maintenance	3-18
Storing Tires After Removal	3-18
Battery Hazard Prevention	3-18
Jump Starting With Jumper Cables	3-19
Jump Starting With Receptacles	3-19
WHEN REPAIRS ARE NECESSARY	3-21
Observe The Following Precautions	3-21
SPECIAL PRECAUTIONS	3-22
Preliminary Procedures Before Welding Or Performing Maintenance	3-22
Engine Shutdown Procedure Before Welding or Performing Maintenance	3-22

Before Welding On The Machine, Obey The Following Guidelines:	3-22
Discharging Control Cabinet Capacitors	3-23
WARNING AND CAUTIONS	
GENERAL INFORMATION	4-2
PIN PLATE	4-2
GRADE SPEED CHART	4-3
MACHINE SAFETY	4-3
KEY SWITCH	4-5
OIL FILL AND CHECK	4-5
HIGH PRESSURES	4-6
BATTERY AND ISOLATION BOX	4-9
HIGH VOLTAGE	4-12
WELDING	4-15
LIFTING POINTS	4-16
EMERGENCY DUMPING	4-17
AUTOMATIC LUBRICATION FILL	4-18
ROPS/FOPS	4-18
TOWING	4-18
LUBRICATION CHART	4-19
OPERATING INSTRUCTIONS	
PREPARING FOR OPERATION	5-2
Safety Is Thinking Ahead	5-2
Walk Around Inspection	5-2
RETRACTABLE LADDER SYSTEM (If Equipped)	5-7
General Information	5-7
Ladder System Description	5-7
Ladder System Operation	5-7
In-Cab Control Panel	5-8
Ground Level Control Box	5-10
Emergency Down Valve	5-11
PRE-SHIFT BRAKE TEST	5-12
Operation	5-12
Service Brake Test	5-13
Parking Brake Test	5-13
Retard System Test	5-14
Brake Test Exit Criteria	5-14
TRUCK OPERATION	5-15
ENGINE START-UP SAFETY	5-15
After Engine Has Started	5-15
MACHINE OPERATION SAFETY PRECAUTIONS	5-16
Machine Operation On The Haul Road	5-17
Starting On A Grade With A Loaded Truck	5-17
Retarding On A Grade	5-17
Passing	5-18
Loading	5-18
Overload Speed Limit Function	5-18
Dumping	5-19
Lowering The Dump Body (When Dumping On Flat Ground)	5-20
Lowering The Dump Body (When Dumping Over A Berm Or Into A Crusher)	5-20
SUDDEN LOSS OF ENGINE POWER OR LOSS OF DRIVE SYSTEM FUNCTION	5-21
EMERGENCY STEERING SYSTEM	5-21
Operation	5-21
Pre-Operation Testing	5-21
Additional Guidelines	5-22

CAMERA SYSTEM (Optional).....	5-23
General.....	5-23
Operation.....	5-23
Camera Monitor.....	5-24
Camera Selection.....	5-24
Auto Backlight Control/Day/Night Settings.....	5-24
Contrast Adjustment.....	5-24
Brightness Adjustment.....	5-24
Saturation Adjustment.....	5-24
Single Scan / Return.....	5-24
Minus.....	5-24
Plus.....	5-24
Enter.....	5-24
FUEL DEPLETION.....	5-25
SAFE PARKING PROCEDURES.....	5-25
NORMAL ENGINE SHUTDOWN PROCEDURE.....	5-25
DISABLED TRUCK CONNECTORS.....	5-27
GENERAL INFORMATION.....	5-27
STEERING AND BRAKE SYSTEM.....	5-27
Hookup.....	5-27
HOIST SYSTEM.....	5-28
Hookup.....	5-28
Dumping Procedure.....	5-29
TOWING.....	5-30
SPECIAL WIRING HARNESS.....	5-30
TOWING PROCEDURE.....	5-30
PAYLOAD METER III SYSTEM.....	5-33
GENERAL INFORMATION.....	5-33
Data Summary.....	5-33
Data Gathering.....	5-33
COMPONENT DESCRIPTION.....	5-33
Operator Display.....	5-33
Operator Switches.....	5-33
Brake Lock Switch.....	5-33
Payload Meter Module.....	5-33
Communications Ports.....	5-34
Load Lights.....	5-34
SCOREBOARD.....	5-34
OPERATOR'S DISPLAY AND SWITCH.....	5-35
Reading the Load Display.....	5-35
Setting the Operator ID.....	5-35
Load and Ton Counter.....	5-35
Viewing Live Sensor Data.....	5-35
Other Display Messages.....	5-36
HAUL CYCLE WARNING FLAGS.....	5-36
Haul Cycle Data.....	5-36
Haul Cycle Warning Flags.....	5-37
PC SOFTWARE OVERVIEW.....	5-39
SYSTEM CONFIGURATION.....	5-39
Connecting to the Payload Meter.....	5-39
Displayed Payload Units.....	5-40
Connection Menu.....	5-40
Configure the Payload Meter.....	5-40
Setting the Date and Time.....	5-40

Setting the Truck Type and Serial Number	5-41
Setting the Gauge Display Units	5-41
Setting the Carryback Alarm Threshold	5-41
Setting the Truck Number	5-42
Setting the Komatsu Distributor	5-42
Setting the Komatsu Customer	5-42
Clean Truck Tare Calibration	5-42
Inclinometer Calibration	5-43
DOWNLOADING PAYLOAD DATA	5-43
UPLOADING DATA	5-44
Creating a Query	5-45
Detailed Multi-Page Report	5-45
Creating Graphs	5-46
DATA ANALYSIS	5-46
Creating a Query	5-46
Payload Detail Screen	5-47
Creating Reports	5-47
Creating Graphs	5-47
Exporting Data	5-48
Importing Data	5-48
Deleting Haul Cycle Records	5-49
Viewing Alarms	5-49
Deleting Alarm Records	5-49
Patent Acknowledgment	5-49
AUTOMATIC LUBRICATION SYSTEM	5-50
GENERAL INFORMATION	5-50
COMPONENT DESCRIPTION	5-50
Filter	5-51
Hydraulic Motor and Pump	5-51
Grease Reservoir	5-51
Pressure Reducing Valve	5-51
Flow Control Valve	5-51
Solenoid Valve	5-52
Vent Valve	5-52
Interface Module	5-52
Injectors	5-53
Grease Pressure Failure Switch	5-54
Relief Valve (unloader valve)	5-54
SYSTEM OPERATION	5-54
Filling The Reservoir	5-54
Priming The Automatic Lubrication System	5-55
PREVENTIVE MAINTENANCE PROCEDURES	5-55
Daily Lubrication System Inspection	5-55
250 Hour Inspection	5-56
1000 Hour Inspection	5-56
SYSTEM CHECKOUT	5-56
General Checks	5-56
Lubrication Cycle Operation	5-57
RESERVE ENGINE OIL SYSTEM (if equipped)	5-58
Operation	5-58
LED Monitor Light	5-59
Changing Oil	5-59
OPERATOR CAB AND CONTROLS	
CAB INTERIOR	6-2

STEERING WHEEL AND CONTROLS	6-3
Horn Button	6-3
Tilt / Telescope Lever	6-3
Multi-Function Turn Signal Switch	6-3
Turn Signal Operation	6-3
High Beam Headlight Operation	6-3
Windshield Wiper Operation	6-3
PEDALS	6-5
Service Brake Pedal	6-5
Dynamic Retard Pedal	6-5
Dynamic Retarding	6-5
Accelerator Pedal	6-5
OVERHEAD PANEL AND DISPLAYS	6-6
Radio Speakers	6-6
Warning Alarm Buzzer	6-6
Cab Radio	6-6
Air Cleaner Vacuum Gauges	6-6
Camera Monitor (Optional)	6-6
CENTER CONSOLE	6-7
Directional Control Lever	6-7
Override/Fault Reset Switch	6-7
Engine Stop Switch	6-7
Window Control Switches	6-8
Hoist Control Lever	6-8
Raising The Dump Body	6-8
Lowering The Dump Body	6-8
Retard Speed Control (RSC) Adjust Dial	6-8
Retard Speed Control (RSC) Switch	6-9
Data Store Button	6-9
12V Auxiliary Power Outlets	6-9
Fire Suppression Control Outlet	6-9
Communication Radio Outlet	6-9
DIAGNOSTIC PORTS	6-10
KOMTRAX Plus	6-10
Interface Module (IM)	6-10
Drive System Or Modular Mining	6-10
Drive System Controller (DSC)	6-10
Engine (CENSE)	6-10
Engine (QUANTUM)	6-10
OPERATOR SEAT (Standard)	6-11
Seat Belts And Tethers	6-11
Adjustment	6-11
PASSENGER SEAT (Standard)	6-13
Seat Belts And Tethers	6-13
Adjustment	6-13
OPERATOR SEAT (Optional)	6-14
Seat Belts And Tethers	6-14
Adjustment	6-14
PASSENGER SEAT (Optional)	6-16
Seat Belts And Tethers	6-16
Adjustment	6-16
HEATER / AIR CONDITIONER CONTROLS	6-17
Fan Speed Control Knob	6-17
Temperature Control Knob	6-17

Air Flow Directional Knob	6-17
Heater/Air Conditioner Vents	6-17
INSTRUMENT PANEL	6-18
Control Symbols	6-18
Heater / Air Conditioner Vents	6-19
Status Indicator Light Panel	6-19
Hydraulic Oil Temperature Gauge	6-19
Coolant Temperature Gauge	6-19
Speedometer/Digital Display	6-19
Fuel Level Gauge	6-19
Drive System Temperature Gauge	6-20
Hazard Warning Light Switch	6-20
Pre-Shift Brake Test Switch	6-20
Wheel Brake Lock Control Switch	6-20
Rest Switch	6-20
Engine Warm Up	6-21
Heated Mirror Switch (Optional)	6-21
Rotating Beacon Light Switch (Optional)	6-21
Camera System (Optional)	6-21
Key Switch	6-21
Warning Light	6-22
Digital Display Contrast Buttons	6-22
Turn Signal Indicators	6-22
Panel Light Dimmer	6-22
Mode Switches	6-22
Fog Light Switch	6-22
Backup Light Switch	6-22
Ladder Light Switch	6-22
Headlight/Panel Illumination Light Switch	6-22
STATUS INDICATOR LIGHT PANEL	6-24
STATUS INDICATOR LIGHT SYMBOLS	6-24
Body Up	6-24
Parking Brake	6-24
Propel System Not Ready	6-24
No Propel	6-25
Engine Warm Up	6-25
Engine Shutdown Delay	6-25
Retarding Grid Dryer	6-25
Wheel Brake Lock	6-25
Traction Control	6-25
Repair Monitor	6-25
Retractable Ladder Down	6-25
Snapshot In Progress	6-25
Pantograph Energized	6-25
Pantograph Up	6-26
Service Brake	6-26
Dynamic Retarding	6-26
No DC Link Voltage	6-26
DIGITAL DISPLAY OPERATION	6-27
Toggling Through Main Level Screens	6-27
Toggling Through Warning Screens	6-28
Entering Payload Meter Data	6-29
KOMTRAX Plus	6-30
KOMTRAX Plus OPERATION	6-30

Basic Precautions	6-30
Interface Module	6-31
DOWNLOADING KOMTRAX Plus DATA	6-31
General Information	6-31
KOMTRAX Plus Downloading Procedure	6-32
Location Of Downloaded Files	6-32
UPLOADING KOMTRAX Plus DATA	6-33
ORBCOMM	6-35
KOMATSU WIRELESS BRIDGE (Optional)	6-36
General Information	6-36
Communication Setup	6-36
Switching To The KWB_SETUP Network	6-36
Setting The Computer Subnet Mask And IP Address	6-37
Setting Up The KWB	6-37
Adding Encryption	6-38
Final Computer Settings	6-38
Downloading Data	6-38
Resetting The KWB	6-39
FUSES AND CIRCUIT BREAKERS	6-40
VEHICLE ELECTRICAL CENTER 89	6-40
VEHICLE ELECTRICAL CENTER 90	6-42
VEHICLE ELECTRICAL CENTER 91	6-44
CIRCUIT BREAKERS	6-46
LUBRICATION AND SERVICE	
GENERAL INFORMATION	7-2
HYDRAULIC SYSTEM SERVICE	7-2
Hydraulic Tank	7-2
Hydraulic Filters	7-3
Hydraulic Pump Drive Shaft	7-3
WHEEL MOTOR SERVICE	7-3
COOLANT LEVEL SERVICE	7-3
Radiator Filling Procedure	7-4
Coolant Specifications	7-4
Unacceptable Practices	7-4
RESERVE ENGINE OIL SERVICE	7-4
Reserve Oil Tank Filling Procedure (Remote Fill)	7-5
Inline Screen	7-5
CAB AIR FILTER SERVICE	7-5
FUEL SYSTEM SERVICE	7-6
Fuel Tank Cleaning	7-6
Fuel Filter Service	7-6
INTAKE AIR CLEANER SERVICE	7-7
FRONT WHEEL SERVICE	7-8
Oil Sampling	7-8
FRONT ENGINE MOUNT TRUNNION SERVICE	7-9
QUICK FILL SERVICE CENTER	7-9
TIE ROD INSPECTION AND TORQUE PROCEDURE	7-9
LUBRICATION CHART	7-11
EVERY 10 HOUR (DAILY) MAINTENANCE	7-12
INITIAL 50 HOUR LUBRICATION AND MAINTENANCE INSPECTION	7-14
INITIAL 100 HOUR LUBRICATION AND MAINTENANCE INSPECTION	7-15
EVERY 250 HOUR LUBRICATION AND MAINTENANCE	7-16
EVERY 500 HOUR LUBRICATION AND MAINTENANCE	7-19
EVERY 1000 HOURS LUBRICATION AND MAINTENANCE	7-21

EVERY 2500 HOUR MAINTENANCE	7-22
EVERY 4000 HOUR MAINTENANCE	7-23
EVERY 5000 HOUR MAINTENANCE	7-24
EVERY 10,000 HOUR MAINTENANCE.....	7-25
EVERY 18,000 HOUR MAINTENANCE.....	7-26
SPECIFICATIONS	
MAJOR COMPONENT DESCRIPTION.....	8-2
Truck And Engine.....	8-2
Traction Alternator.....	8-2
Main Control Cabinet.....	8-2
AC Induction Traction Motorized Wheels.....	8-2
Suspension.....	8-2
Operator's Cab.....	8-2
Power Steering	8-2
Dynamic Retarding Grid.....	8-2
Brake System	8-2
SPECIFICATIONS	8-3
Engine.....	8-3
AC Electric Drive System.....	8-3
Dynamic Retarding.....	8-3
Battery Electric System.....	8-3
Service Capacities.....	8-3
Hydraulic Systems.....	8-3
Service Brakes.....	8-4
Parking Brake	8-4
Steering.....	8-4
Standard Dump Body Capacities And Dimensions.....	8-4
Tires.....	8-4
Weight Distribution	8-4
Overall Truck Dimensions	8-5
RADIO	
RADIO OPERATION	9-2
General Radio Receiver Functions.....	9-2
Turning On The Power	9-2
One-Hour Timer.....	9-2
Finding A Station	9-2
Setting The Presets	9-3
Adjusting The Receiver Audio Settings.....	9-3
Display.....	9-4
Adjusting The Receiver Menu Settings.....	9-4
Front Auxiliary Mode	9-4
Rear Auxiliary Mode	9-4
CLOCK	9-5
Clock/Alarm Button	9-5
Setting The Clock	9-5
Setting The Alarm.....	9-5
Turning The Alarm Off.....	9-6
Activating Snooze.....	9-6
CD PLAYER	9-7
Playing A Compact Disc.....	9-7
Playing An MP3 DISC.....	9-7
CD Mode Displays.....	9-7
Button Functions	9-7
MP3 PLAYER	9-9

Playing MP3 Files.....	9-9
MP3 Mode Displays.....	9-9
Button Functions	9-9
iPod®	9-11
Playing An iPod®	9-11
iPod® Mode Displays	9-11
Button Functions	9-11
WEATHERBAND (USA Only)	9-13
TROUBLESHOOTING	9-14
Tips About the Audio System.....	9-14
Understanding Radio Reception	9-14
Care Of The Compact Discs.....	9-14

INTRODUCTION

FOREWORD F.1-0000051238

This manual is written for use by the operator and/or the service technician and is designed to help these persons become fully knowledgeable of the truck and all its systems in order to keep it operating safely and efficiently.

All operators and maintenance personnel must read and understand the materials in this manual before operating the truck or performing maintenance and/or operational checks on the truck. All safety notices, warnings and cautions are to be understood and followed when operating or accomplishing repairs on the truck.

The illustrations used in this manual are typical of the component shown and may not be an exact reproduction of what is found on the truck.

A product identification plate is located on the frame in front of the right side front wheel and designates the truck model number, product identification number (vehicle serial number), and maximum GVW (Gross Vehicle Weight) rating.

The KOMATSU truck model designation consists of three numbers and one letter (i.e. 830E).

The three numbers represent the basic truck model.

The letter "M", when present, designates a Mechanical drive system;

The letter "E", when present, designates an Electrical wheel motor drive system.

The product identification number (vehicle serial number) contains information which will identify the original manufacturing bill of material for this unit. This complete number will be necessary for proper ordering of many service parts and/or warranty consideration.

The GVW is what determines the load on the drive train, frame, tires, and other components. The vehicle design and application guidelines are sensitive to the total maximum GVW. GVW is TOTAL WEIGHT: the Empty Vehicle Weight + the fuel & lubricants + the payload.

To determine allowable payload: Service all lubricants for proper level and fill fuel tank of empty truck (which includes all accessories, body liners, tailgates, etc.) and then weigh truck. Record this value and subtract from the GVW rating. The result is the allowable payload.

NOTE: *Accumulations of mud, frozen material, etc. become a part of the GVW and reduces allowable payload. To maximize payload and to keep from exceeding the GVW rating, these accumulations should be removed as often as practical.*

Exceeding the allowable payload will reduce expected life of truck components.

 DANGER

- **Unsafe use of this machine may cause serious injury or death. Operators and maintenance personnel must read and understand this manual before operating or maintaining this machine.**
- **This manual should be kept in or near the machine for reference, and periodically reviewed by all personnel who will come into contact with it.**

ABOUT THIS MANUAL F.2-0000051237

This manual is written for use by the operator and/or the service technician. It is designed to help these persons learn how to operate the truck and its systems in order to keep it operating safely and efficiently. All operators and maintenance personnel must read and understand the materials in this manual before operating the truck or performing maintenance and/or operational checks on the truck. All safety notices, warnings, and cautions should be understood and followed when operating or performing repairs on the truck.

The front cover of this manual includes a form number. This form number should be referenced when ordering additional copies of this manual, or for any other correspondence regarding the coverage in this manual.

Direct all inquiries to:
Komatsu America Corp
Literature Services
100 Komatsu Dr.
Cartersville, GA 30121
United States of America
Phone: 678-721-3310
Email: KACTEHPUBS@KOMATSUNA.COM

If there is ever a question regarding the information in a particular section, refer to the manual form number, and use the address shown above to correspond.

The illustrations used in this manual are typical of the component shown and may not be an *exact* reproduction of what is found on the truck.

This manual shows dimensioning of U.S. standard and metric (SI) units throughout. All references to “right”, “left”, “front”, or “rear” are made with respect to the operator’s normal seated position, unless specifically stated otherwise.

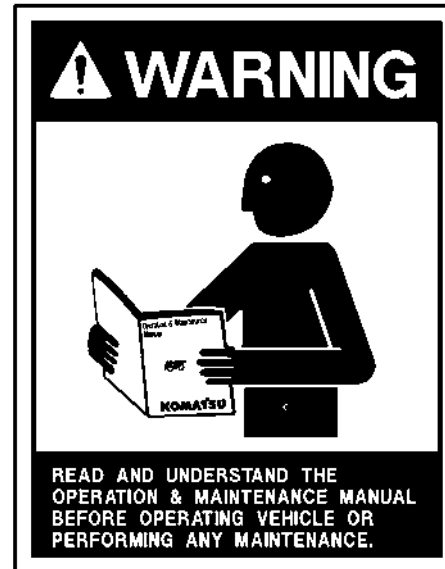
When assembly instructions are provided without references to torque values, standard torque values should be assumed. Standard torque requirements are shown in torque charts in the following section and in the General Information section of the truck service manual. Individual torques when provided in the text are in bold face type, such as **135 N·m (100 ft lb)**. All torque specifications have $\pm 10\%$ tolerance unless otherwise specified.

WARNINGS F.3-0000051687

This material is proprietary to Komatsu America Corp (KAC), and is not to be reproduced, used, or disclosed except in accordance with written authorization from KAC.

It is the policy of the company to improve products whenever it is possible and practical to do so. The company reserves the right to make changes or add improvements at any time without incurring any obligation to install such changes on products sold previously.

Because of continuous research and development, periodic revisions may be made to this publication. Customers should contact their local Komatsu distributor for information on the latest revision.

**! DANGER**

- Unsafe use of this machine may cause serious injury or death. Operators and maintenance personnel must read and understand this manual before operating or maintaining this machine.
- This manual should be kept in or near the machine for reference, and periodically reviewed by all personnel who will come into contact with it.

California Proposition 65 Warnings**! WARNING**

- Diesel engine exhaust, some of its constituents, and certain vehicle components contain or emit chemicals known to the State of California to cause cancer, birth defects or other reproductive harm.

! WARNING

- Battery posts, terminals and related accessories contain lead and lead compounds, chemicals known to the State of California to cause cancer, birth defects or other reproductive harm. Wash hands after handling.

! WARNING

- Mercury and mercury compounds are known to the State of California to cause developmental problems. This machine may be equipped with optional HID lamps which contain mercury. There is no risk of exposure unless the lamps are broken. However, the lamps must be reused, recycled or properly disposed of in accordance with Local, State and Federal Laws at the end of their useful lives.

ALERTS D.2-0000051268

83403

The ALERT symbol is used with the following signal words:

- DANGER
- WARNING
- CAUTION

This symbol is used to alert the reader of hazards that can arise from improper operating and maintenance practices.

! DANGER

- **DANGER** identifies a specific potential hazard WHICH WILL RESULT in either INJURY OR DEATH if proper precautions are not taken.

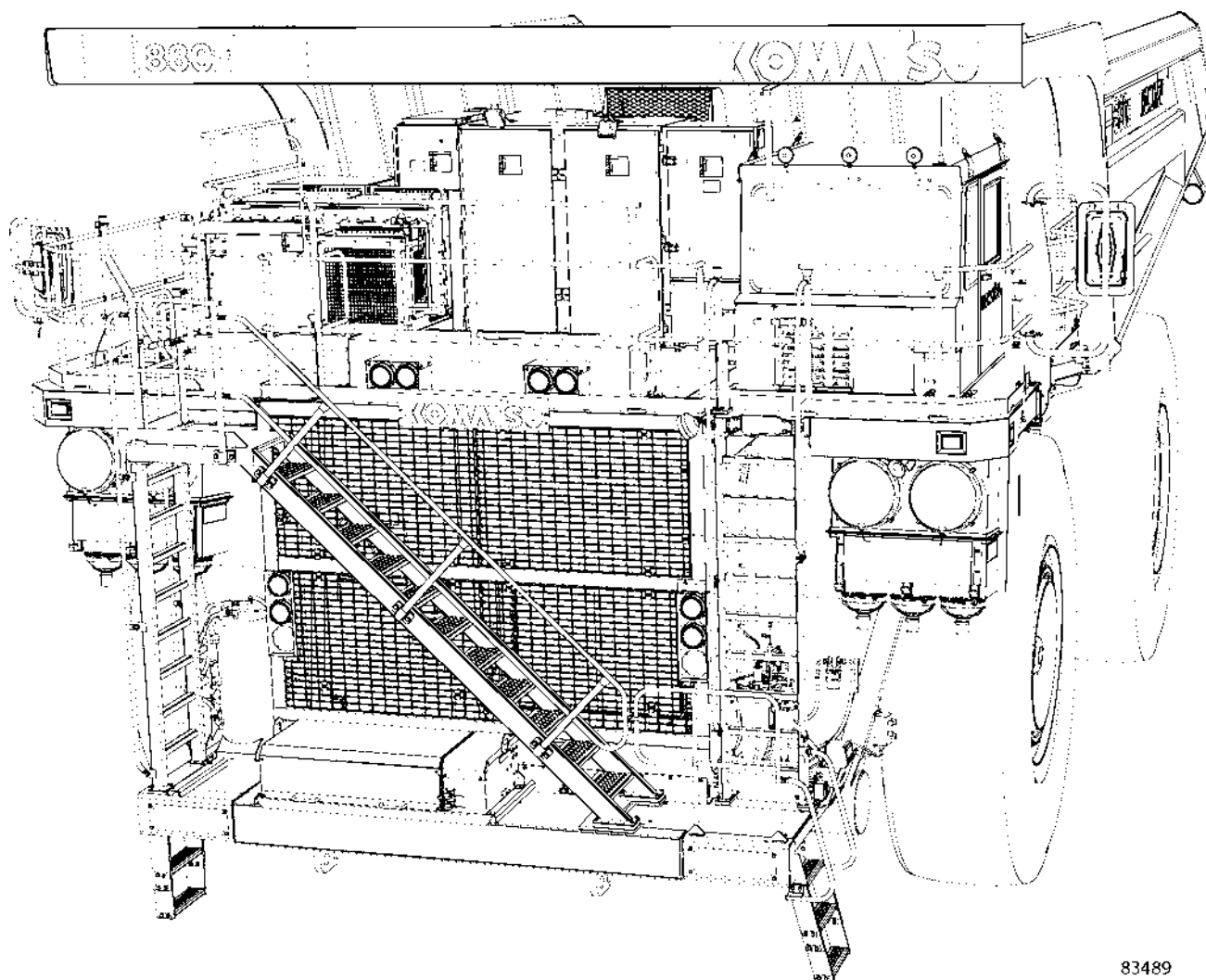
! WARNING

- **WARNING** identifies a specific potential hazard WHICH COULD RESULT in either INJURY OR DEATH if proper precautions are not taken.

! CAUTION

- **CAUTION** is used for general reminders of proper safety practices OR to direct the reader's attention to avoid unsafe or improper practices which may result in damage to the machine.

TRUCK MODEL ILLUSTRATION A.5-0000054230



83489

NON-OEM PARTS IN CRITICAL SYSTEMS C.3-0000051235

For safety reasons, Komatsu America Corp. strongly recommends against the use of non-OEM replacement parts in critical systems of all Komatsu equipment. Critical systems include but are not limited to steering, braking and operator safety systems

Replacement parts manufactured and supplied by unauthorized sources may not be designed, manufactured or assembled to Komatsu's design specifications; accordingly, use of such parts may compromise the safe operation of Komatsu products and place the operator and others in danger should the part fail.

Komatsu is also aware of repair companies that will rework or modify an OEM part for reuse in critical systems. Komatsu does not generally authorize such repairs or modifications for the same reasons as noted above.

Use of non-OEM parts places full responsibility for the safe performance of the Komatsu product on the supplier and user. Komatsu will not in any case accept responsibility for the failure or performance of non-OEM parts in its products, including any damages or personal injury resulting from such use.

STANDARD CHARTS AND TABLES

WARNING

Read and follow all safety precautions. Failure to do so may result in serious injury or death.

This safety section also contains precautions for optional equipment and attachments.

GENERAL INFORMATION E.2-0000051234

This manual provides dual dimensioning for many specifications. Metric units are specified first, with U.S. standard units in parentheses. When torque values are not specified in the assembly instructions contained in this manual, use the standard torque value for the hardware being used. Standard value torque tables are contained in this chapter for metric and SAE hardware.

This truck is assembled with both metric and SAE (U.S.) hardware. Reference the correct table when determining the proper torque value.

⚠ CAUTION

- **References throughout the manual to standard torques or other standard values will be to one of the following tables. Do not use standard values to replace specific torque values in assembly instructions.**

For values not shown in any of the charts or tables, standard conversion factors for most commonly used measurements are provided in the following tables.

Effect Of Special Lubricants

The Komatsu engineering department does not recommend the use of special friction-reducing lubricants, such as Copper Coat, Never-Seez®, and other similar products, on the threads of standard fasteners where standard torque values are applied. The use of special friction-reducing lubricants will significantly alter the clamping force being applied to fasteners during the tightening process.

If special friction-reducing lubricants are used with the standard torque values listed in this chapter, excessive stress and possible breakage of the fasteners may result.

Where the torque tables specify "Lubricated Threads" for the standard torque values listed, these standard torque values are to be used with simple lithium base chassis grease (multi-purpose EP NLGI) or a rust-preventive grease (see below) on the threads and seats unless specified otherwise.

NOTE: *Ensure the threads of fasteners and tapped holes are free of burrs and other imperfections before assembling.*

Suggested Sources For Rust Preventive Grease

NOTE: *This list represents the current engineering approved sources for use in Komatsu manufacture. It is not exclusive. Other products may meet the same specifications of this list.*

- American Anti-Rust Grease #3-X from Standard Oil Company (also American Oil Co.)
- Gulf NoRust #3 from Gulf Oil Company.
- Mobilarma 355, Product No. 66705 from Mobil Oil Corporation.
- Rust Ban 326 from Humble Oil Company.
- Rustolene B Grease from Sinclair Oil Co.
- Rust Preventive Grease - Code 312 from the Southwest Grease and Oil Company.

STANDARD TIGHTENING TORQUES

F.6-0000051233

Class 10.9 Capscrews And Class 10 Nuts

The following specifications apply to required assembly torques for all metric class 10.9 finished hexagon head capscrews and class 10 nuts.

DO NOT lubricate capscrew threads and seats during assembly. These specifications are based on the following hardware:

- All capscrews
- All nuts
- All hardened washers being phosphate and oil coated

NOTE: If zinc-plated hardware is used, each piece must be lubricated with simple lithium base chassis grease (multi-purpose EP NLGI) or a rust preventive grease (see list, this page) to achieve the same clamping forces provided below.

NOTE: Torques are calculated to give a clamping force of approximately 75% of proof load. The maximum torque tolerance shall be within $\pm 10\%$ of the torque value shown.

In the following table under "Capscrew Size", the first number represents the shank diameter (mm). The second number represents thread pitch in millimeters.

Example: M20 x 2.25

M20 = shank diameter (20 mm)

2.25 = thread pitch in millimeters

Table 2-1

Standard Tightening Torque for Metric Class 10.9 Capscrews & Class 10 Nuts			
			
Capscrew Size	Torque N·m	Torque ft·lb	Torque kg·m
M6 x 1	12	9	1.22
M8 x 1.25	30	22	3.06
M10 x 1.5	55	40	5.61
M12 x 1.75	95	70	9.69
M14 x 2	155	114	15.81
M16 x 2	240	177	24.48
M20 x 2.25	465	343	47.43
M24 x 3	800	590	81.6
M30 x 3.5	1600	1180	163.2
M36 x 4	2750	2028	280.5

SAE Grade 5 And Grade 8 Capscrews

The following specifications apply to required assembly torques for all grade 5 and grade 8 capscrews.

NOTE: Capscrew threads and seats shall be lubricated when assembled. Unless instructions specifically recommend otherwise, these standard torque values are to be used with simple lithium base chassis grease (multi-purpose EP NLGI) or a rust-preventive grease (see list, previous page) on the threads.

NOTE: Torques are calculated to give a clamping force of approximately 75% of proof load. The maximum torque tolerance shall be within $\pm 10\%$ of the torque value shown.

In the following table under Capscrew Size, the first number represents the shank diameter (in.). The second number represents threads per inch.

Example: 7/16 - 20

7/16 = shank diameter (7/16 inch (0.438 inch))

20 = threads per inch

Table 2-2

Standard Tightening Torque for SAE Hex Head Capscrew And Nut Assembly						
 						
Cap-screw Size	Torque Grade 5			Torque Grade 8		
	N·m	ft·lb	kg·m	N·m	ft·lb	kg·m
1/4-20	9.5	7	0.97	13.6	10	1.38
1/4-28	10.8	8	1.11	14.9	11	1.52
5/16-18	20.3	15	2.07	28	21	2.90
5/16-24	22	16	2.21	30	22	3.04
3/8-16	34	25	3.46	47	35	4.84
3/8-24	41	30	4.15	54	40	5.5
7/16-14	54	40	5.5	79	58	8.0
7/16-20	61	45	6.2	84	62	8.57
1/2-13	88	65	9	122	90	12.4
1/2-20	95	70	9.7	129	95	13.1
9/16-12	122	90	12.4	169	125	17.3

Standard Tightening Torque for SAE Hex Head Capscrew And Nut Assembly						
 						
Cap-screw Size	Torque Grade 5			Torque Grade 8		
	N·m	ft·lb	kg·m	N·m	ft·lb	kg·m
9/16-18	129	95	13.1	183	135	18.7
5/8-11	169	125	17.3	237	175	24.2
5/8-18	183	135	18.7	258	190	26.2
3/4-10	298	220	30.4	420	310	42.8
3/4-16	319	235	32.5	454	335	46.3
7/8-9	475	350	48.4	678	500	69.2
7/8-14	508	375	51.9	719	530	73.3
1.0-8	712	525	72.6	1017	750	103.7
1.0-12	759	560	77.4	1071	790	109.3
1.0-14	773	570	78.8	1085	800	110.6
1 1/8-7	881	650	89.9	1424	1050	145
1 1/8-12	949	700	96.8	1546	1140	158
1 1/4-7	1234	910	125.9	2007	1480	205
1 1/4-12	1322	975	134.8	2142	1580	219
1 3/8-6	1627	1200	166	2630	1940	268
1 3/8-12	1776	1310	181	2874	2120	293
1 1/2-6	2142	1580	219	3471	2560	354
1 1/2-12	2305	1700	235	3756	2770	383

SAE Grade 9 Capscrews

The following specifications apply to required assembly torques for SAE Hex Head and 12-point, grade 9 (170,000 psi minimum tensile), capscrews.

NOTE: Capscrew threads and seats shall be lubricated when assembled. Unless instructions specifically recommend otherwise, these standard torque values are to be used with simple lithium base chassis grease (multi-purpose EP NLGI) or a rust-preventive grease (see list, this page) on the threads.

NOTE: Torques are calculated to give a clamping force of approximately 75% of proof load. The maximum torque tolerance shall be within $\pm 10\%$ of the torque value shown.

Table 2-3

Standard Tightening Torque for SAE Hex Head and 12-Point, Grade 9 Capscrews			
  Hex Head 12-Point			
Capscrew Size ¹	Torque N·m	Torque ft·lb	Torque kg·m
0.250 - 20	16	12	1.7
0.312 - 18	33	24	3.3
0.375 - 16	57	42	5.8
0.438 - 14	95	70	9.7
0.500 - 13	142	105	14.5
0.562 - 12	203	150	20.7
0.625 - 11	278	205	28.3
0.750 - 10	488	360	49.7
0.875 - 9	780	575	79.4
1.000 - 8	1166	860	119
1.000 - 12	1240	915	126
1.125 - 7	1670	1230	170

Standard Tightening Torque for SAE Hex Head and 12-Point, Grade 9 Capscrews

  Hex Head 12-Point			
Capscrew Size ¹	Torque N·m	Torque ft·lb	Torque kg·m
1.125 - 12	1800	1330	184
1.250 - 7	2325	1715	237
1.250 - 12	2495	1840	254
1.375 - 6	3080	2270	313
1.375 - 12	3355	2475	342
1.500 - 6	4040	2980	411
1.500 - 12	4375	3225	445

T-Bolt Clamps

Table 2-4

Tightening Torque For T-Bolt Type Hose Clamp (SAE J1508 Type TB)			
			
Thread Size	Band Width	Newton meters (N·m)	Inch Pounds (in·lb)
0.25-28 UNF	19.05 mm (0.75 in.)	8.5 $\pm$ 0.6	75 $\pm$ 5

¹ Shank Diameter (in) — Threads per inch

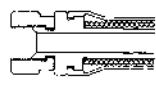
Split Flange Clamps

Table 2-5

Tightening Torque For Split Flange Clamp Bolts				
				
Capscrew Thread Diameter (mm)	Width Across Flat (mm)	Newton meters (N·m) ± 10%	Foot Pounds (ft·lb) ± 10%	Kilogram meters (kg·m) ± 10%
10	14	66	48	6.7
12	17	112	83	11.5
16	22	279	206	28.5

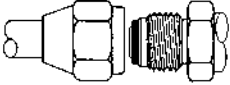
Flared Tube And Hose Fittings

Table 2-6

Tightening Torque For Flared Tube And Hose Fittings				
				
Thread Diameter of Nut (mm)	Width Across Flat (mm)	Newton meters (N·m) ± 10%	Foot Pounds (ft·lb) ± 10%	Kilogram meters (kg·m) ± 10%
14	19	25	18	2.5
18	24	50	36	5
22	27	80	58	8
24	32	140	101	14
30	36	175	130	18
33	41	195	145	20
36	46	245	180	25
42	55	295	215	30

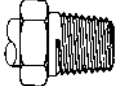
JIC 37° Swivel Nuts

Table 2-7

Torque Chart For JIC 37° Swivel Nuts With Or Without O-ring Seals				
				
Size Code	Tube Size (OD)	Threads UNF-2B	Newton meters (N·m)	Foot Pounds (ft·lb)
– 2	0.125	0.312 – 24	5 ± 1	4 ± 1
– 3	0.188	0.375 – 24	11 ± 4	8 ± 3
– 4	0.250	0.438 – 20	16 ± 4	12 ± 3
– 5	0.312	0.500 – 20	20 ± 4	15 ± 3
– 6	0.375	0.562 – 18	24 ± 7	18 ± 5
– 8	0.500	0.750 – 16	40 ± 7	30 ± 5
– 10	0.625	0.875 – 14	54 ± 7	40 ± 5
– 12	0.750	1.062 – 12	75 ± 7	55 ± 5
– 14	0.875	1.188 – 12	88 ± 7	65 ± 5
– 16	1.000	1.312 – 12	108 ± 7	80 ± 5
– 20	1.250	1.625 – 12	136 ± 14	100 ± 10
– 24	1.500	1.875 – 12	163 ± 14	120 ± 10
– 32	2.000	2.500 – 12	312 ± 27	230 ± 20

Pipe Thread Fittings

Table 2-8

Torque Chart For Pipe Thread Fittings					
					
Size Code	Pipe Thread Size	With Sealant N·m	With Sealant ft·lb	Without Sealant N·m	Without Sealant ft·lb
– 2	0.125 – 27	20 ± 4	15 ± 3	27 ± 7	20 ± 5
– 4	0.250 – 18	27 ± 7	20 ± 5	34 ± 7	25 ± 5
– 6	0.375 – 18	34 ± 7	25 ± 5	48 ± 7	35 ± 5
– 8	0.500 – 14	48 ± 7	35 ± 5	61 ± 7	45 ± 5
– 12	0.750 – 14	61 ± 7	45 ± 5	75 ± 7	55 ± 5
– 16	1.000 – 11.50	75 ± 7	55 ± 5	88 ± 7	65 ± 5
– 20	1.250 – 11.50	95 ± 7	70 ± 5	108 ± 7	80 ± 5
– 24	1.500 – 11.50	108 ± 7	80 ± 5	129 ± 14	95 ± 10
– 32	2.000 – 11.50	129 ± 14	95 ± 10	163 ± 14	120 ± 10

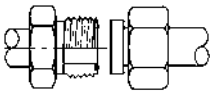
O-Ring Boss Fittings

Table 2-9

Torque Chart For O-ring Boss Fittings				
				
Size Code	Tube Size (OD)	Threads UNF-2B	Newton meters (N·m)	Foot Pounds (ft·lb)
– 2	0.125	0.312 – 24	4 ± 3	4 ± 2
– 3	0.188	0.375 – 24	7 ± 3	5 ± 2
– 4	0.250	0.438 – 20	11 ± 4	8 ± 3
– 5	0.312	0.500 – 20	14 ± 4	10 ± 3
– 6	0.375	0.562 – 18	18 ± 4	13 ± 3
– 8	0.500	0.750 – 16	33 ± 7	24 ± 5
– 10	0.625	0.875 – 14	43 ± 7	32 ± 5
– 12	0.750	1.062 – 12	65 ± 7	48 ± 5
– 14	0.875	1.188 – 12	73 ± 7	54 ± 5
– 16	1.000	1.312 – 12	98 ± 7	72 ± 5
– 20	1.250	1.625 – 12	109 ± 7	80 ± 5
– 24	1.500	1.875 – 12	109 ± 7	80 ± 5
– 32	2.000	2.500 – 12	130 ± 14	96 ± 10

O-Ring Face Seal Fittings

Table 2-10

Torque Chart For O-ring Face Seal Fittings				
				
Size Code	Tube Size (O.D.)	Threads UNF-2B	Newton meters (N·m)	Foot Pounds (ft·lb)
– 4	0.250	0.438 – 20	15 ± 1	11 ± 1
– 6	0.375	0.562 – 18	24 ± 3	18 ± 2
– 8	0.500	0.750 – 16	48 ± 5	35 ± 4
– 10	0.625	0.875 – 14	69 ± 7	51 ± 5
– 12	0.750	1.062 – 12	96 ± 10	71 ± 7
– 16	1.000	1.312 – 12	133 ± 8	98 ± 6
– 20	1.250	1.625 – 12	179 ± 10	132 ± 7
– 24	1.500	1.875 – 12	224 ± 20	165 ± 15

This as a preview PDF file from **best-manuals.com**



Download full PDF manual at **best-manuals.com**